Jhih-Jhong (Johnny) Lin

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Education

Master of Science in Computer Science and Information Engineering — Taipei, Taiwan
National Taiwan University — *Aug. 2026 - Present*

Bachelor of Science in Computer Science and Information Engineering — Taipei, Taiwan
National Taiwan University — *Aug. 2022 - Jul. 2026*

Cumulative GPA: 4.23/4.30 | **Class Rank**: 7/150 | **2 Dean's List Awards**

Relevant Courses: Algorithm Design and Analysis (A+), Cryptography and Network Security (A+), Theoretical Aspects of Modern Cryptography (A+), Compiler Design (A+), Formal Methods (A), Types and Programming Languages (A+), Operating Systems (A+), Virtual Machines (A), Computer Architecture (A+)

Experience

Product Development Intern — New Taipei, Taiwan
Synology, Inc. — *Sept. 2026 - Present*

- Automate the conversion of 100+ manual QA test specifications into executable tests with LLM agents.

Visiting Research Student — Irvine, California
Department of Informatics, University of California, Irvine | *Advisor: Prof.* ***Joshua Garcia*** — *Jun. 2026 - Aug. 2026*

- Surveyed 16 agentic LLM security benchmarks and evaluated three frameworks for cybersecurity education.
- Developed a reproducible evaluation workflow and analyzed benchmark behavior across locally hosted LLMs.

Research Intern / Project Assistant — Taipei, Taiwan
Institute of Information Science (IIS), Academia Sinica | *Advisor: Prof.* ***Bow-Yaw Wang*** — *Jul. 2025 - Sept. 2026*

- Extended CryptoLine using OCaml to model floating-point instructions and verify them with the Bitwuzla SMT solver.
- Developed abstract interpretation techniques for verification of floating-point cryptographic programs.

Undergraduate Researcher — Taipei, Taiwan
Network Security Lab (NSLab), CSIE, National Taiwan University | *Advisor: Prof.* ***Hsu-Chun Hsiao*** — *Sept. 2024 - Jul. 2026*

- Investigated integration of directed fuzzers with recurring vulnerability detection tools.
- Currently exploring formal methods, including static analysis and cryptographic protocol verification.

Teaching Assistant, Introduction to Computational Logic — Taipei, Taiwan
Department of Computer Science and Information Engineering, National Taiwan University — *Fall 2025*

Network Administrator (Member of NASA Team) — Taipei, Taiwan
Department of Computer Science and Information Engineering, National Taiwan University — *Sept. 2023 - Jun. 2025*

- Maintained Cisco and Juniper switch infrastructure serving 600+ faculty members and students.
- Built automated monitoring and backup workflows using Prometheus and Grafana for newly deployed switches.

Teaching Assistant, Network Administration and System Administration — Taipei, Taiwan
Department of Computer Science and Information Engineering, National Taiwan University — *Spring 2025*

Skills

Languages: Mandarin (native); English (advanced, TOEFL iBT 5.5/6)
Programming: C, Python, OCaml, Bash; working knowledge of C++, Java, and Haskell
Software: Linux, Git, LaTeX, Docker, CPAchecker, CryptoLine, Compiler (flex, yacc), Fuzzing (AFL++)

Selected Projects

Evaluating Agentic LLM Security Benchmarks for Cybersecurity Education: A Reproducible Study
Artifact | Slides | UCInspire Research Project

MerkleDir.sh: A shell script that creates Merkle trees from directories and generates/verifies inclusion proofs
Code (Bash) | Developed as a TA for Network Administration and System Administration

CTF Challenge Design: Cryptography, Fuzzing, and DNS Cache Poisoning
DNS Cache Poisoning | Cryptography and Fuzzing | Developed as a TA for Network Administration and System Administration

An Evaluation on Possibilities of RCE and Defense Mechanism in Electron-Based Applications
Co-authored with: Kang-Yu Fan, Tsan-Huan Hsu, Yi-Ting Lo
Project Report | Cryptography and Network Security Course Project

Grey-Box Symbolic Execution with Likely-Invariant Generation in CPAchecker
Code (Java) | Project Report | Formal Methods Course Project

Terminal-Based P2P and Centralized Chatroom Implementation
Co-authored with: Wen-Chun Kao
Code (C++) | Computer Networks Course Project